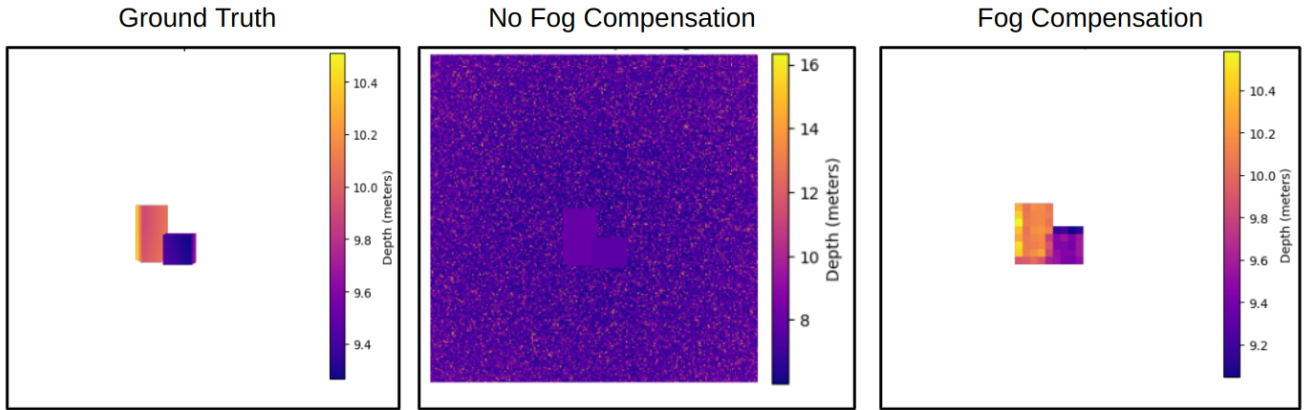


Histogram-Based Defogging of SPAD LiDAR

Nathan Ludlow
Carnegie Mellon University
Robotics Institute
nludlow@andrew.cmu.edu



Abstract

Adverse weather conditions, particularly fog, pose significant challenges to the reliability of LiDAR-based perception systems in autonomous vehicles. In this work, we present a novel physics-informed defogging method for Single Photon Avalanche Diode (SPAD) LiDAR that reconstructs accurate depth maps in foggy conditions without relying on learning-based models. Our approach extends histogram-based defogging techniques to handle spatially varying fog by dividing the scene into spatial bins and estimating fog parameters locally using an Expectation-Maximization (EM) algorithm. We then recover object depths using a Generalized Likelihood Ratio Test (GLRT). Simulations in Mitsuba using transient rendering show that our method achieves low RMSE under moderate fog and remains effective even in dense fog conditions. The method runs in real time, is compatible with existing SPAD LiDAR hardware, and demonstrates robust performance without the need for large training datasets, making it well-suited for safety-critical applications in autonomous vehicle systems.

1. Introduction

Almost all current autonomous vehicles rely heavily on Single Photon Avalanche Diode (SPAD) LiDAR to accurately map their surroundings in real time [10]. As a central component of the vehicle’s perception stack, SPAD LiDAR provides high-resolution spatial information critical for safe navigation. However, the reliability of these systems hinges on the quality and robustness of sensor measurements under varying environmental conditions.

One of the most significant challenges to LiDAR perception is the presence of adverse weather—particularly fog. Fog can severely degrade LiDAR performance through attenuation and backscattering effects [2, 3, 17, 19]. While recent advances in SPAD LiDAR have produced finer temporal resolution and improved signal fidelity, these sensors continue to struggle in dense fog. This is especially problematic for 905nm LiDAR systems, which dominates the automotive market due to cost and manufacturability. Unlike 1550nm LiDAR, which better penetrates fog droplets, 905nm systems are inherently more susceptible to scattering, limiting their range and accuracy in poor visibility.

Given that most deployed and upcoming autonomous vehicles use 905nm LiDAR, the inability to perceive reliably

in fog poses a critical safety risk. Fog can emerge unexpectedly, and autonomous systems must be equipped to adapt in real time. Despite this urgent need, few practical and deployable solutions currently exist to mitigate fog effects in LiDAR measurements.

Given these issues, this work proposes a novel physics-informed defogging framework that improves the reconstruction of LiDAR depth maps under foggy conditions, without relying on deep learning. Two primary approaches currently dominate the defogging landscape:

1. **Histogram-based techniques** estimate and subtract fog profiles from raw LiDAR histograms based on physical models of photon propagation in fog. These methods typically assume a uniform fog distribution across the scene [13].
2. **Deep learning approaches** aim to learn an inverse mapping from fog-degraded input to clean depth or point cloud outputs. Some process raw ToF histograms, while others work directly on the point cloud representation. [1]

Histogram-based methods are fast and interpretable, making them suitable for real-time applications. However, they suffer when fog is spatially non-uniform, and often focus only on object detection rather than depth recovery. Deep learning methods, while powerful and more flexible to scene variation, are generally too computationally expensive for real-time use in high-frame-rate SPAD systems. Furthermore, the potential risk of false depth outputs makes them less desirable in safety-critical settings.

To address the shortcomings of existing approaches, this work introduces a locally adaptive histogram-based defogging method capable of reconstructing accurate depth maps in non-uniform fog conditions. The method operates by first grouping neighboring pixels into small spatial bins, then fitting a fog attenuation model to each bin using a statistical regression process. By leveraging localized estimates of fog density, the method can more effectively isolate object returns and estimate depth, even in scenes where fog distribution varies.

The proposed approach offers several key benefits:

- It enables real-time, physically grounded fog compensation compatible with existing SPAD LiDAR hardware.
- It reconstructs usable depth maps rather than binary object presence, providing richer scene understanding.
- It handles spatially varying fog without requiring learning-based models, increasing robustness.

Improving LiDAR reconstruction under foggy conditions enhances not only the reliability of autonomous vehicles but also broadens the operational envelope for other robotics platforms that rely on LiDAR for navigation and perception. By advancing real-time, physics-based defogging, this work contributes a critical capability toward safer and more adaptable autonomous systems.

2. Related Works

Fog and other adverse weather conditions significantly degrade LiDAR performance, creating a major challenge for autonomous vehicles that rely on accurate LiDAR-based environment reconstruction for navigation. A range of solutions has been proposed to tackle this issue, including statistical methods, physics-based models, and deep learning approaches. Each method has its strengths and weaknesses, but all emphasize the need to improve LiDAR perception in challenging weather conditions.

Automotive manufacturers and researchers have been working to create datasets that enable simulated testing of LiDAR in adverse weather. These datasets are critical for quickly testing and evaluating perception algorithms under difficult conditions. The Canadian Adverse Driving Conditions (CADC) dataset [11] provides annotated LiDAR data collected in winter driving environments, giving researchers a way to test their models in real-world conditions. Northrop et al. further explored how well LiDAR performs in various adverse weather conditions using both real-world data and simulations [9]. Their results suggest that while current LiDAR systems generally perform acceptably in most adverse weather, there is still a significant drop in performance compared to normal conditions. Because of the adequate performance in adverse weather, manufacturers have been able to deploy vehicles, but improving LiDAR measurements in these scenarios could make autonomous vehicles far safer and more reliable, especially given the safety-critical nature of these systems.

Given the importance of high-quality sensor data for autonomous vehicles, researchers have also explored hardware adaptations to enhance LiDAR performance in poor weather. Kutilla et al. tested 905 nm and 1550 nm LiDAR systems under controlled fog and rain conditions, showing that longer wavelengths penetrate fog more effectively due to reduced scattering [8]. Similarly, Heinzler et al. examined how weather affects automotive LiDAR sensors and proposed classification techniques to dynamically adjust sensor processing [4]. 1550 nm LiDAR systems show greatly improved performance over 905 nm LiDAR in adverse weather but due to their increased costs, they have not been widely adopted by existing automotive manufacturers.

In contrast to hardware advancements, many researchers have explored leveraging physics-based principles to improve LiDAR scans in real time. Histogram-based defogging techniques, such as those proposed by Sang and Tsai [13, 14], utilize statistical estimation (e.g., Expectation Maximization) to model fog profiles and remove their effects from LiDAR histograms. These methods assume uniform fog conditions, which limits their applicability in real-world scenarios where fog density varies spatially. Similarly, Satat et al. [16] employ statistical regression on LiDAR input to distinguish between light reflections from ob-

jects and light scattering in fog. Tobin et al. extend this concept by specifically focusing on detecting moving objects in fog [18]. A related U.S. patent further enhances this approach by dynamically estimating fog concentration using blank channel information, though it still struggles with non-uniform fog distributions [15]. These studies highlight the importance of integrating physical principles into LiDAR defogging techniques, which have demonstrated strong performance in removing uniform fog effects but all face challenges with non-uniform fog conditions and other complex weather phenomena.

Other physics-based methods have also gained traction. Hahner et al. developed a physically valid fog simulation algorithm that transforms clear-weather LiDAR point clouds into foggy counterparts using the Beer-Lambert law and Mie scattering theory [3]. Yang et al. extended this work by incorporating additional weather simulations in the CARLA simulator, including rain, as well as splash and spray effects from vehicle wheels, to test the approach under various adverse conditions [19]. Similar to statistical modeling methods, these approaches struggle with non-uniform conditions.

Deep learning methods offer an alternative by training neural networks to estimate fog or other weather effects and reconstruct corrupted point clouds. For example, Gated Recurrent Units (GRUs) have been used to transform degraded LiDAR data into accurate reconstructions [13]. However, these approaches require large training datasets and are computationally intensive at runtime. Kilic et al. address this limitation by introducing a hybrid Monte Carlo-based simulation framework (LISA) to augment training datasets with realistic fog effects modeled using Mie scattering theory [7]. Kempen et al. leverage temporal information in sequential LiDAR frames to enhance object detection under adverse weather conditions [6]. Although these deep learning approaches have shown effectiveness in reconstructing LiDAR scans in adverse weather, their high computational costs limit their feasibility for real-time sensor reconstructions in autonomous vehicles. This project builds upon these works by addressing the limitations of histogram-based defogging methods in non-uniform fog conditions. By dividing histograms into spatial cells and processing each cell independently, this approach aims to model varying fog densities across scenes effectively.

This project builds upon these works by addressing the limitations of histogram-based defogging methods in non-uniform fog conditions. By dividing histograms into spatial cells and processing each cell independently, this approach aims to model varying fog densities across scenes effectively. This method leverages insights from physics-based simulations and experimental studies to improve reconstruction accuracy while maintaining computational efficiency.

3. Method

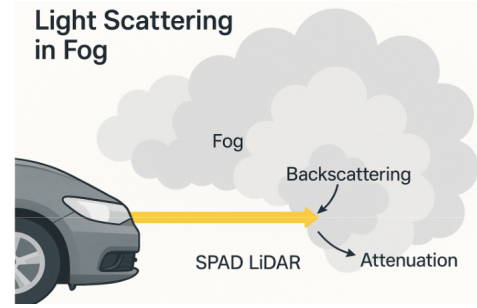


Figure 1. The two main issues which cause problems for SPAD LiDARs in fog are backscattering and attenuation. backscattering causes a large number of false positives in the sensor, while attenuation makes it harder to detect actual objects in front of you.

SPAD LiDAR systems, despite their high temporal resolution and single-photon sensitivity, are significantly impacted by adverse weather conditions. In particular, fog introduces two dominant challenges: *backscattering*, where photons are scattered back toward the sensor before reaching the object, and *attenuation*, which reduces the number of photons returning from the scene. These effects distort the transient histogram, making depth estimation unreliable. See Figure 1

To address these challenges, we extend the physically grounded fog modeling framework proposed by Sang et al. [14], which models the fog return using an exponentially decaying modified gamma function which fits fog parameters using Expectation-Maximization (EM). Unlike their approach, which assumes spatially uniform fog, our method operates locally on bins of pixels, estimating fog parameters per spatial bin to handle non-uniform fog distributions across the scene.

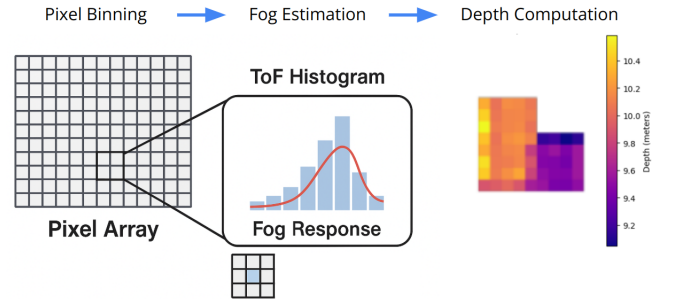


Figure 2. The pipeline for our proposed method. We start by binning pixels into groups, then estimating the fog response in each bin, then finally computing depth for each bin after removing the fog effects.

Our pipeline proceeds in three main stages (See Figure 2):

pixel binning, *fog estimation*, and *depth computation*. The purpose of pixel binning is to reduce the variance in the observed transient histograms by averaging spatially neighboring pixels. This increases the signal-to-noise ratio (SNR) and stabilizes parameter estimation in low-photon conditions. In the fog estimation step, we fit a physically grounded fog model to the averaged histogram of each bin using an EM algorithm, recovering the extinction coefficient, fog onset distance, and fog scale factor. Finally, in the depth computation step, we use a likelihood ratio test to identify the most probable object depth in the defogged histogram, accepting a detection if the peak is both statistically significant and spatially consistent. Each of these stages is described in detail below.

Pixel Binning

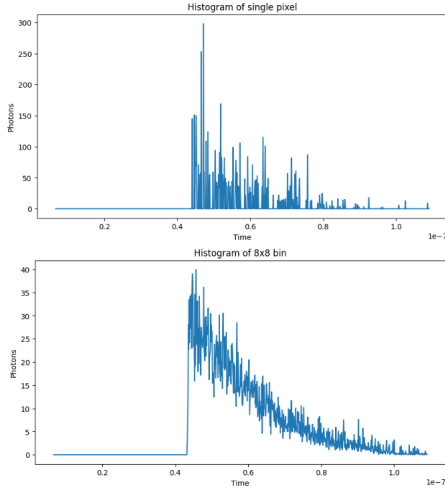


Figure 3. The difference in histograms between a single pixel, and an averages histogram in an 8x8 bin. It can be seen clearly that the exponential decay function of the fog is much stronger in the 8x8 bin, as opposed to the single pixel.

The signal quality in SPAD LiDAR data is often limited by photon shot noise, particularly under adverse weather conditions. To mitigate this, we perform spatial binning of the transient image prior to fog estimation. Specifically, in this work we divide the 2D transient image into non-overlapping 8×8 pixel blocks. In the next step when we estimate fog parameters we estimate parameters per bin of pixels by and computing the mean histogram within each block. The effects of this binning can be seen in Figure 3 where the exponential decay signal from the fog can be clearly seen observed when averaging the pixel histograms as opposed a reasonably unclear decay function from a single pixel. In the single pixel histogram it is hard to differentiate what part of the fog signal is noise, as opposed to the scattering effects of the fog and as a result would be hard to

fit a fog function to.

Binning is done as defined in Equation 1.

$$\bar{Y}_{i,j}(R) = \frac{1}{|B_{i,j}|} \sum_{(x,y) \in B_{i,j}} Y_{x,y}(R), \quad (1)$$

where $B_{i,j}$ denotes the set of pixel coordinates in the (i,j) -th bin, and $|B_{i,j}|$ is the number of pixels in the bin. This averaging process greatly reduces variance and produces a higher signal-to-noise ratio (SNR) histogram for subsequent EM-based fog parameter estimation.

After fog estimation and depth computation, the results from each bin are upsampled to the original image resolution. We assign each pixel in a 8×8 block the corresponding depth value estimated for that bin. While this introduces a resolution tradeoff, it greatly enhances the robustness of parameter estimation under noisy conditions, and is well-suited for fog scenarios where fine-grained resolution is less critical than reliable depth recovery.

Fog Model Fitting

Each pixel in a SPAD LiDAR system accumulates a histogram $y_i(R)$, where R denotes the time-of-flight bin (i.e., range). In our work we take this $y_i(R)$ and find the mean of all the histograms of a bin which gives us $Y_i(R)$ representing the histogram of a single bin. Under foggy conditions, the received photon counts are modeled as:

$$Y_i(R) = Y_{\text{fog},i}(R) + Y_{\text{obj},i}(R) + \lambda_w, \quad (2)$$

where:

- $Y_{\text{fog},i}(R)$ is the fog-induced return,
- $Y_{\text{obj},i}(R)$ is the return from the target,
- λ_w is the background noise level.

Following the LiDAR model in [14], the fog return $Y_{\text{fog},i}(R)$ is approximated as $\lambda_{\text{fog}}(R; K_{\text{fog}}, \mu_{\text{ext}}, R_1)$ which is expressed as:

$$\lambda_{\text{fog}}(R; K_{\text{fog}}, \mu_{\text{ext}}, R_1) = K_{\text{fog}} \frac{\mu_{\text{ext}}}{R^2} e^{-2\mu_{\text{ext}}R} u(R - R_1), \quad (3)$$

where K_{fog} is a scaling factor, μ_{ext} is the extinction coefficient, R_1 is the fog onset distance, and $u(\cdot)$ is the Heaviside step function. We use this same model to represent fog in our work.

The $1/R^2$ term models spherical spreading loss, while the exponential term models two-way attenuation along the outgoing and return paths of the light. This equation is similar to a gamma function and fits the model of light fall off and scattering of fog found through numerous external studies.

If we ignore noise, we can formulate an optimization function to find our fog parameters as:

$$\min \sum_{R=1}^m (Y(R) - \lambda_{fog}(R; K_{fog}, \mu_{ext}, R_1))^2 = \quad (4)$$

$$\min \sum_{R=1}^m \left(Y(R) - K_{fog} \frac{\mu_{ext}}{R^2} e^{-2\mu_{ext}R} u(R - R_1) \right)^2 \quad (5)$$

Here, m is the total number of histogram bins in the averages transient response of a single bin. The goal of the optimization is to estimate the fog parameters K_{fog} , μ_{ext} , and R_1 , which represent the fog intensity scaling factor, the extinction coefficient, and the onset range of the fog, respectively. The inclusion of R_1 allows the model to handle scenes where the vehicle has not yet entered the fog but is observing it from a distance. This formulation assumes that the aggregate photon return from fog is significantly stronger than that from scene objects within each spatial bin. As such, the fog model is fit to the combined sensor response. While this means the model does not exclusively fit the fog component, the assumption that fog returns dominate over object returns in most conditions makes the approximation both practical and robust.

2. Fog Model Parameter Fitting We solve for the optimal fog profile in each bin b by minimizing the squared error between observed and modeled photon counts:

$$\min_{\mu_{ext}, R_1} \sum_{i \in b} \sum_{R=1}^m (Y_i(R) - \lambda_{y,i}(R; \mu_{ext}, R_1, R_{2,i}))^2. \quad (6)$$

Due to difficulties optimizing this with more advanced methods, we optimize this using an Estimation Maximization (EM) procedure. This allows us to optimize the parameters despite the discontinuities in this for the model function. After running our optimization, we get a set of estimated fog profile parameters for each bin in the image.

1. Initialization of EM

- K_{fog} is set to one.
- $\mu_{ext}^{(b)}$ is initialized to zero.
- $R_1^{(b)}$ is initialized as the first time bin where the patch-averaged histogram exceeds a basic CFAR threshold.

Depth Computation

After removing the estimated fog contribution from each pixel's histogram, we apply a Generalized Likelihood Ratio Test (GLRT) to determine the most probable depth at which an object return occurs. For each range bin R , we evaluate the log-likelihood ratio between two hypotheses: the null hypothesis H_0 , where the histogram contains only fog and background noise, and the alternative hypothesis $H_1(R)$,

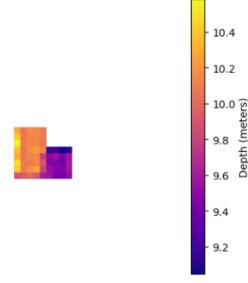


Figure 4. Sample return of a complete depth reconstruction using our method.

where an object return is present at bin R in addition to fog and noise. The log-likelihood ratio is computed as:

$$\log \Lambda(R) = Y(R) \cdot \log \left(\frac{\lambda_1}{\lambda_0} \right) - (\lambda_1 - \lambda_0), \quad (7)$$

where $\lambda_0 = \lambda_{fog}(R) + \lambda_w$ is the expected photon rate under H_0 , and $\lambda_1 = \lambda_0 + \lambda_{obj}$ is the rate under $H_1(R)$, assuming a fixed object return strength λ_{obj} . The bin with the maximum log-likelihood ratio is selected as the depth estimate, provided that it exceeds a predefined detection threshold and exhibits sufficient contrast relative to neighboring bins. This procedure enables robust depth detection by comparing out sensor return from a bin to the expected return of a fog-only response to determine if an object is in the response as well.

After performing this full pipeline, we end up with a complete depth map for each bin the SPAD sensor. We then convert this back to the original depth map space by repeating the value of the bin in each pixel used to create the bin, and doing this we generate a complete depth map. A complete reconstructed depth map can be seen in Figure 4.

4. Results

All experiments were conducted in simulation using the Mitsuba renderer [5], with the Mitransient package [12] used to generate transient LiDAR data. To evaluate the performance of our fog compensation method, we tested it across a range of fog densities, varying two key parameters: the extinction coefficient μ_{ext} and the scattering coefficient μ_s , both defined within the renderer.

The extinction coefficient μ_{ext} represents the fraction of light attenuated before reaching the sensor, while the scattering coefficient μ_s defines the fraction of light scattered by the medium. Both parameters are normalized between 0 and 1.

We simulated a simplified braking scenario by placing a rectangular object (1.8 m tall, 0.4 m wide) 40 meters in front of the sensor—approximating the size of a pedestrian at a

critical distance corresponding to the stopping range of a vehicle traveling at 50 mph. The entire scene was enveloped in fog, and we systematically varied μ_{ext} and μ_s to observe the method’s performance under different fog conditions.

We report the average root mean square error (RMSE) of the estimated depth relative to ground truth in Table 1, demonstrating the effectiveness and limitations of the fog compensation algorithm across a range of visibility conditions. Each set of fog parameterizations had 5 trials run and the average RSME is reported in the table. In all trials a small amount of Gaussian noise was added to the sensor array, and photons were sampled from a Poisson distribution to more accurately model real world sensor effects.

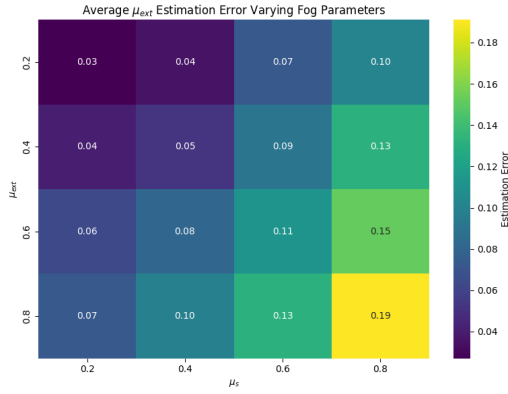


Figure 5. This shows the estimation error of μ_{ext} across the scene. The estimated μ_{ext} of all bins was averaged and compared with the true simulator value of μ_{ext} to find the absolute value of the error of the parameter estimation for each fog parameterization.

We also report the accuracy of our parameter estimation for μ_{ext} to demonstrate the efficacy of our parameter estimation approach in Figure 5. We compare the estimated fog attenuation coefficient with the true value to show how accurately we can model the fog in our approach using our EM algorithm with out fog model.

Due to the considerable computational time required to render transients across the full parameter grid, this braking scenario was the only experiment we were able to complete within the timeframe.

5. Discussion

The results presented in Table 1 demonstrate that our fog compensation method maintains relatively low RMSE values under light to moderate fog conditions. As expected, the estimation error increases with higher values of both the extinction coefficient μ_{ext} and the scattering coefficient μ_s . Notably, even in severe fog conditions ($\mu_{\text{ext}} = 0.8$, $\mu_s = 0.8$), the method is able to recover the object’s location within approximately 3 meters of its true depth—providing reasonably accurate information for a 40m range in a criti-

Table 1. Results of the braking test. Average Root Mean Square Error (RMSE) of the estimated depth map after applying the fog compensation algorithm across 5 trials per parameterization is shown in the table. A range of scattering and attenuation coefficients were used to simulate different fog conditions. In all trials, the object was placed at 40 meters from the camera to mimic a realistic driving scenario requiring rapid deceleration.

	$\mu_s = 0.2$	$\mu_s = 0.4$	$\mu_s = 0.6$	$\mu_s = 0.8$
$\mu_{\text{ext}} = 0.2$	0.523 m	0.671 m	1.023 m	1.537 m
$\mu_{\text{ext}} = 0.4$	0.602 m	0.814 m	1.205 m	1.879 m
$\mu_{\text{ext}} = 0.6$	0.636 m	0.893 m	1.681 m	2.366 m
$\mu_{\text{ext}} = 0.8$	0.659 m	0.968 m	2.033 m	2.982 m

cal driving scenarios.

As shown in Figure 5, the parameter estimation error also increases with both μ_{ext} and μ_s , indicating that denser fog degrades the accuracy of fog model recovery. This trend aligns with the depth estimation results, reinforcing the observation that higher scattering and attenuation reduce signal quality and model fidelity. Nevertheless, the method maintains low estimation error (well below 0.1) under mild fog conditions and continues to produce usable results even with imperfect parameter estimates. These findings suggest that this method exhibits a high degree of resilience across a range of fog conditions.

Overall, the proposed physics-based histogram defogging framework demonstrates itself to be a highly viable approach for enhancing autonomous vehicle perception in foggy environments. The method runs in real-time and is grounded in physical modeling, making it compatible with existing SPAD LiDAR hardware. It can reconstruct accurate depth maps under various fog conditions with only minor loss of resolution and does not rely on learning-based models, while still handling spatially varying fog in a robust way.

We acknowledge that this evaluation was limited to a single controlled scene due to the high computational cost of rendering full transient histograms with fog. As such, there are several directions for future work. First, evaluating the method on scenes with spatially varying fog would help verify its ability to generalize beyond uniform conditions, particularly given the binning structure of the algorithm. Second, testing at varying ranges and on scenes where fog only partially occludes the field of view would further validate the method’s robustness. Finally, implementation on real SPAD hardware would provide critical validation of the method’s real-world applicability; however, these were not feasible to complete for this work due to time and budget.

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